SMS Spam Detection Using Machine Learning

1. Abstract

This project uses Machine Learning to classify SMS messages as spam or ham (normal messages). The dataset contains 5,572 messages. Text was converted into numerical features using TF-IDF, and a Multinomial Naive Bayes model was trained. The model achieved 96.05% accuracy on the test dataset.

2. Introduction

SMS messages are commonly used for communication, but some messages contain unwanted promotions or suspicious offers. This project explores how Machine Learning can help identify such messages automatically.

3. Problem Statement

The aim is to build a model that can classify a text message into one of two categories: spam or ham.

4. Objectives

• To load and understand an SMS dataset.

• To convert message text into numerical features using TF-IDF.

• To train a Multinomial Naive Bayes classifier.

• To evaluate the model using accuracy, a classification report, and a confusion matrix.

5. Dataset Description

The dataset contains 5,572 rows and two columns: **label** and **message**. There are 4,825 ham messages and 747 spam messages. No missing values were found.

6. Methodology

• Loaded the SMS dataset using Pandas.

• Checked the dataset shape, columns, missing values, and label counts.

• Split the data into 80% training data (4,457 messages) and 20% testing data (1,115 messages), using stratification.

• Used TF-IDF to convert the text into numerical features. The training vocabulary contained 7,668 features.

• Trained a Multinomial Naive Bayes model using the training data.

7. Results and Evaluation

The model achieved **96.05% accuracy** on the test dataset. The classification report showed an F1-score of 0.98 for ham and 0.83 for spam. Spam recall was 0.70, which means the model did not identify every spam message.

Actual category	Predicted ham	Predicted spam
Ham	966	0
Spam	44	105

The confusion matrix shows that 966 ham messages and 105 spam messages were classified correctly. Another 44 spam messages were incorrectly classified as ham.

8. Sample Prediction

A sample message, “Congratulations! You have won a free prize. Call now!”, was classified as SPAM by the model.

9. Limitations and Future Scope

This project is a beginner-level demonstration and was evaluated on one test split. Some spam messages were missed. In future, the model could be improved by testing other algorithms, tuning settings, and using more varied messages. The prediction should not be treated as a guarantee that a real message is safe or unsafe.

10. Conclusion

In this project, I built an SMS classification model using TF-IDF and Multinomial Naive Bayes. The model achieved 96.05% accuracy on the test dataset. I also used a classification report and confusion matrix to understand its performance. This project helped me learn the basic steps of preparing text data, training a model, and evaluating its predictions.

Note: This report describes an educational project. Results may differ on new or real-world SMS messages.